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LAB ASSIGNMENT-01

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REGISTER NUMBER : 23MID0061

COURSE TITLE : ADVANCED PYTHON PROGRAMMING

COURSE CODE : CSI3007

COURSE FACULTY : SHARMILA BANU K

LAB SLOT : L31+L32+L49+L50

SUBMISSION DATE : 16-07-2026

TASK -1 :

Comparison of ANSI Encoding and Unicode Encoding.

AIM :

To understand the difference between ANSI encoding and Unicode encoding by saving a Chinese character in different encoding formats using Notepad and observing the results.

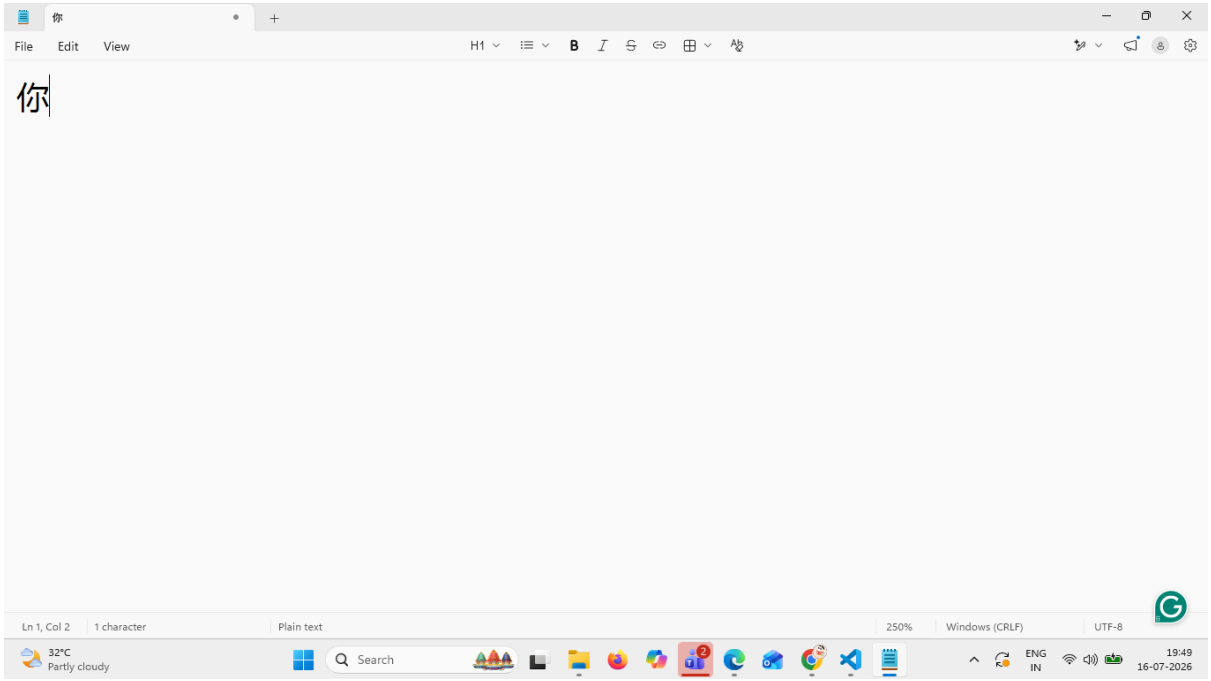
1.ANSI Encoding

ANSI encoding is a traditional character encoding system that represents characters using a **single-byte (8-bit)** code. It can store a maximum of **256 characters** in a single code page. ANSI is based on code pages, where each code page is designed to support a specific language or group of languages.

Since ANSI has a limited number of character codes, it cannot represent all the characters used in different languages. If a character is not available in the selected code page, it is replaced with a placeholder symbol such as **?**.

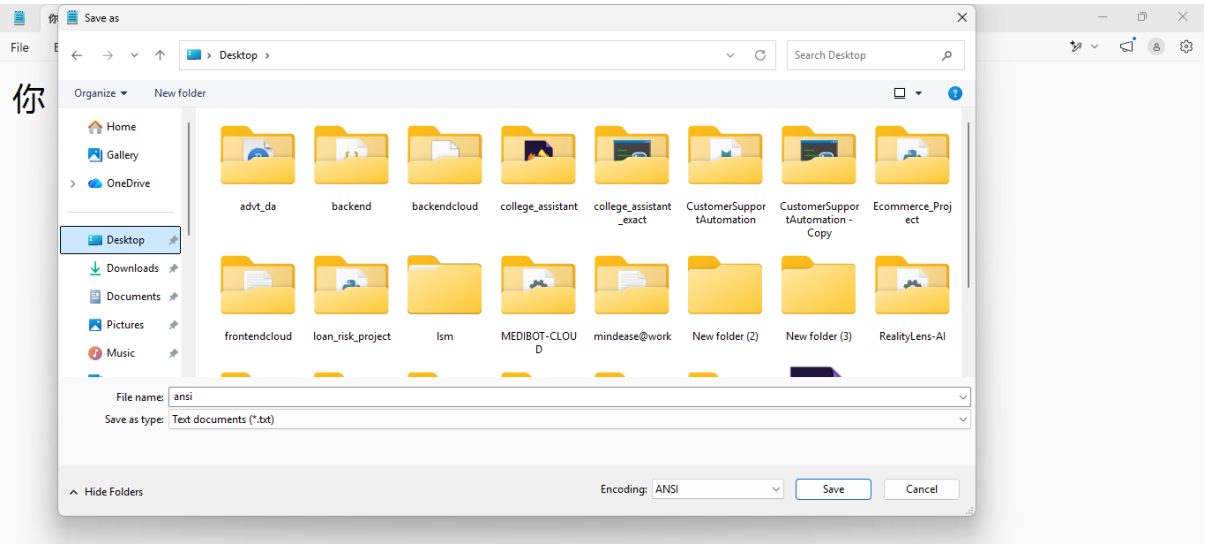
STEPS :

1.Enter a chinese character in the notepad :



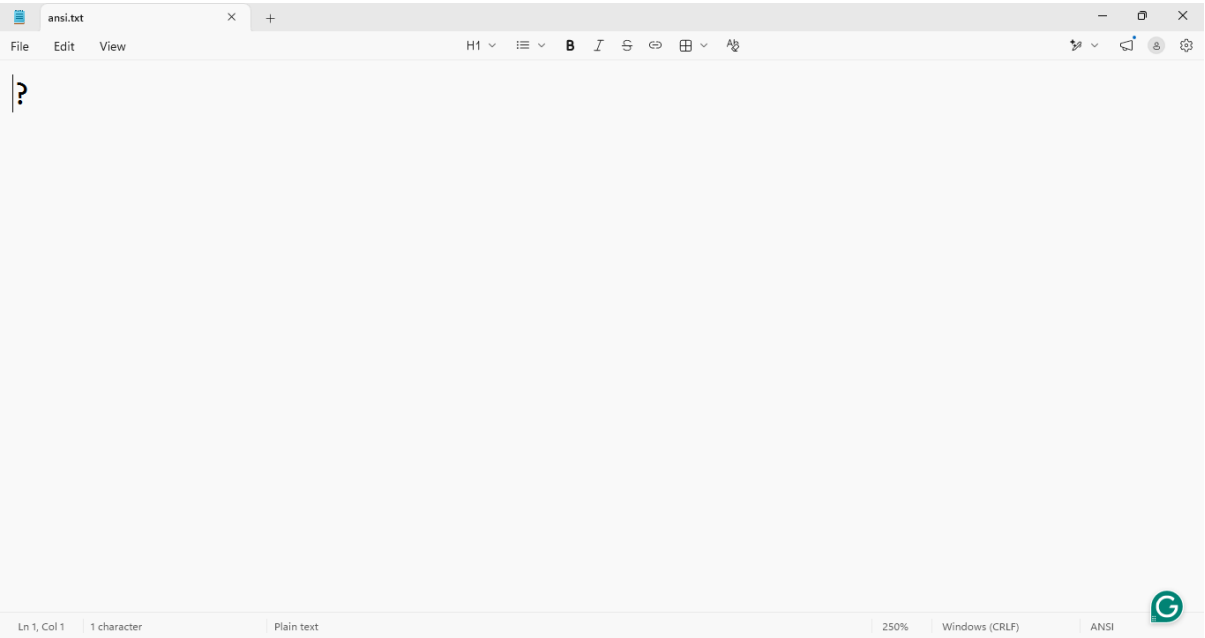
2. Save the File Using ANSI Encoding.

- Click File → Save As.
- Enter a file name (eg : ansi.txt).
- In the Encoding drop-down list, select ANSI.
- Click Save.



3.Reopen the File

Close the file and open **ansi.txt** again.The original Chinese character is replaced by a **?**



Observation :ANSI supports only the characters available in its selected code page. Since the Chinese character "你" is not included in most English or Indian ANSI code pages.Notepad cannot assign it a valid character code. As a result, it replaces the character with a question mark (?) when the file is saved.

2. Unicode Encoding

Unicode is a universal character encoding standard that assigns a unique code point to every character from almost all languages in the world. It supports over 149,000 characters**,** including letters, numbers, symbols, emojis, and multilingual text. Unicode is commonly implemented using encoding formats such as UTF-8**,** UTF-16**,** and UTF-32**.**

Example :

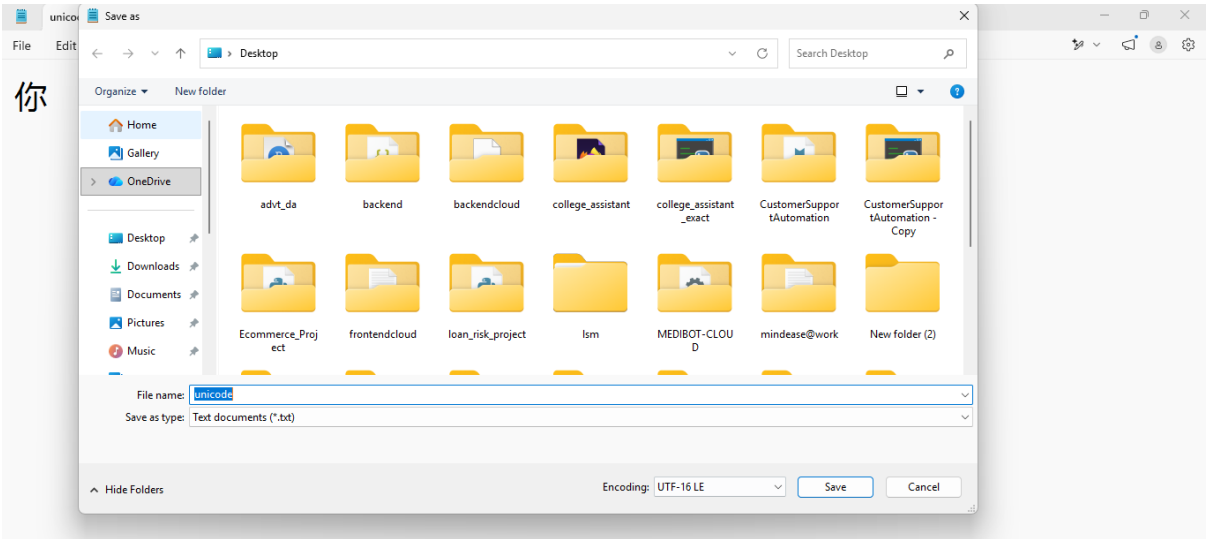
Character Unicode Code Point Binary Representation (UTF-8)

A U+0041 01000001

STEPS :

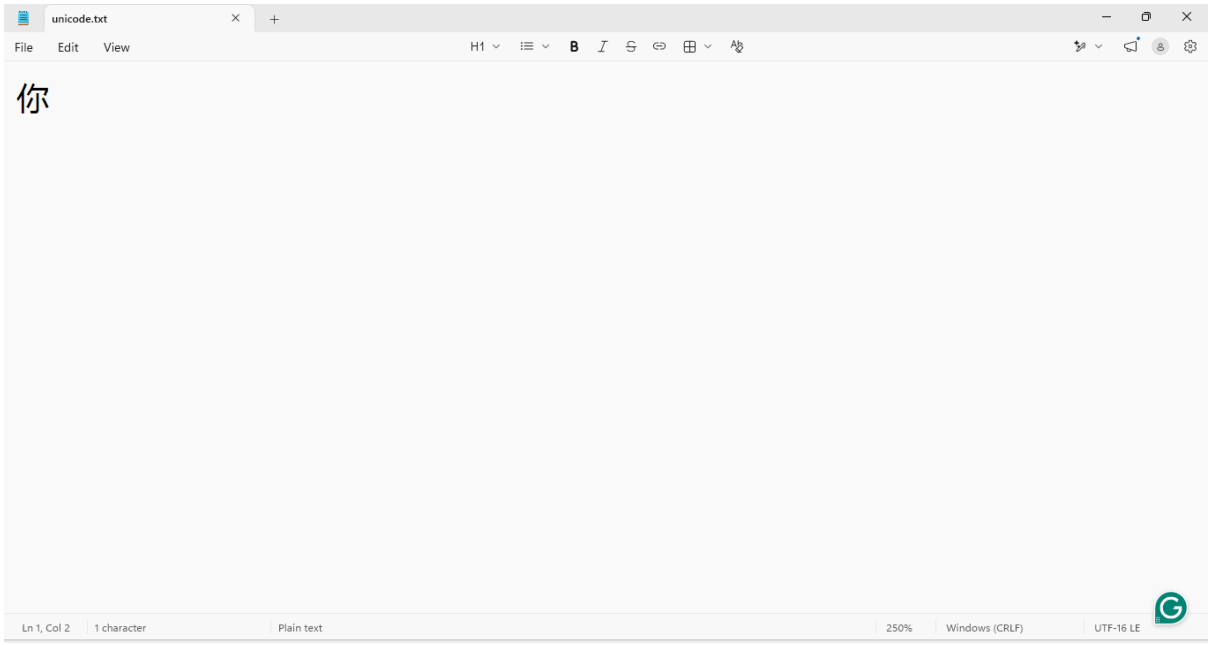
1. Type the Chinese character into Notepad:

- Click **File → Save As**.
- Enter a file name (for example, unicode.txt**).**
- In the **Encoding** drop-down list, select **UTF-8** (or **Unicode (UTF-16 LE)** in older versions of Notepad).
- Click **Save**.



2.Reopen the File :

Close the file and open **unicode.txt** again.The Chinese character 你 is displayed correctly without any changes.



Observation :

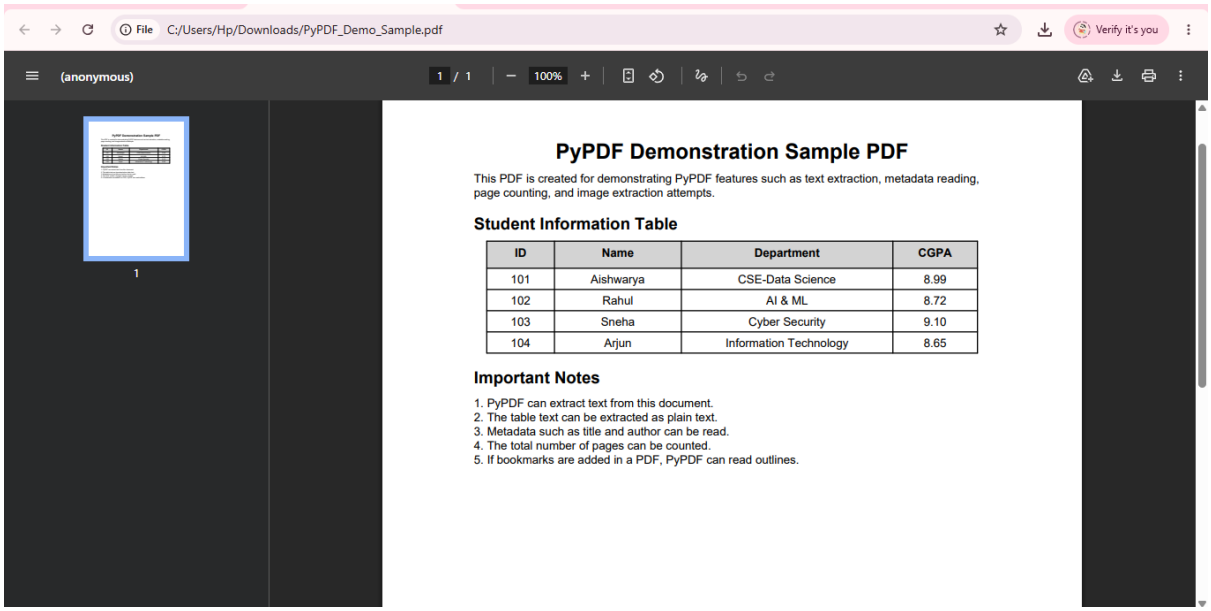
Unicode assigns a unique code point to every character. Since the Chinese character 你 has the code point **U+4F60**, it can be encoded and stored correctly using UTF-8 or UTF-16. Therefore, when the file is reopened, the original character is displayed without any changes.

Therefore,from this experiment we can conclude **ANSI** and **Unicode** use different methods to store text. ANSI supports only a limited number of characters, so unsupported characters like the Chinese character 你 are replaced with a **?**. Unicode supports characters from almost all languages, so it stores and displays them correctly. Therefore, Unicode is more reliable and is widely used in modern computers and applications.

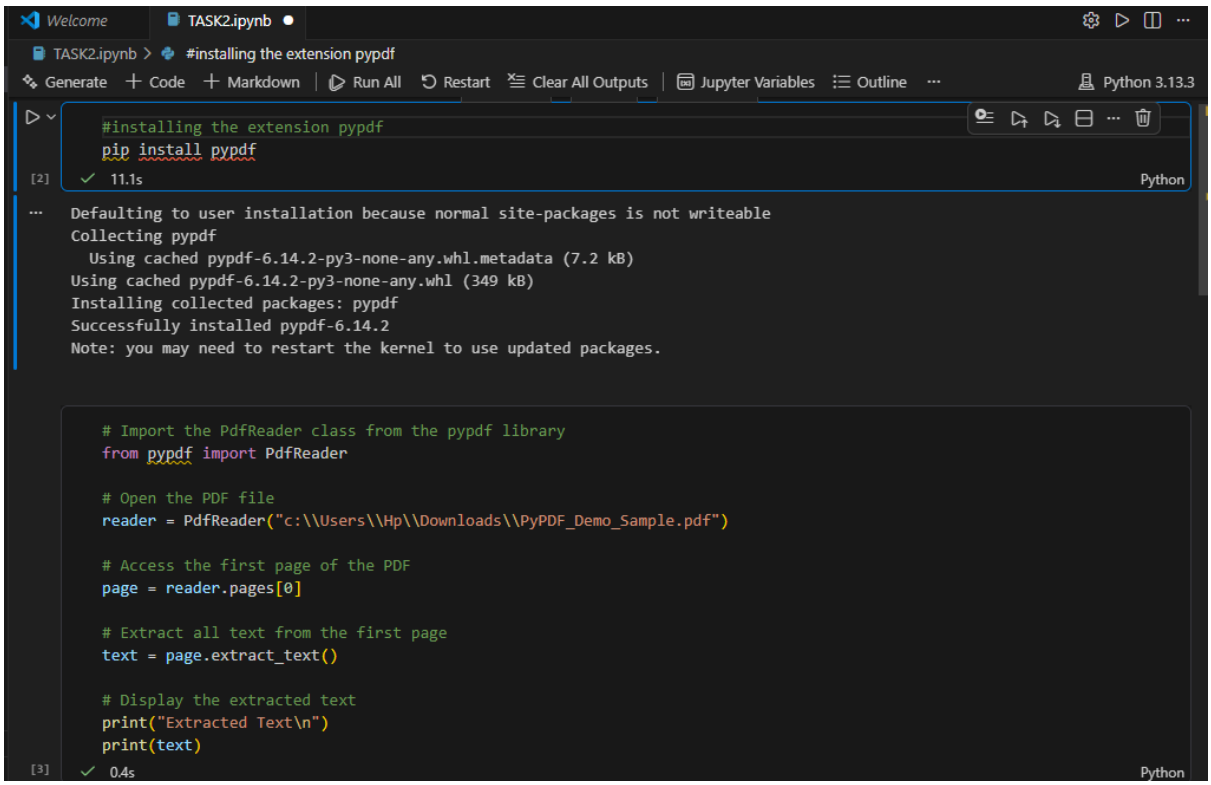
TASK -2 : Exploring python extension for file formats.

1.PyPDF

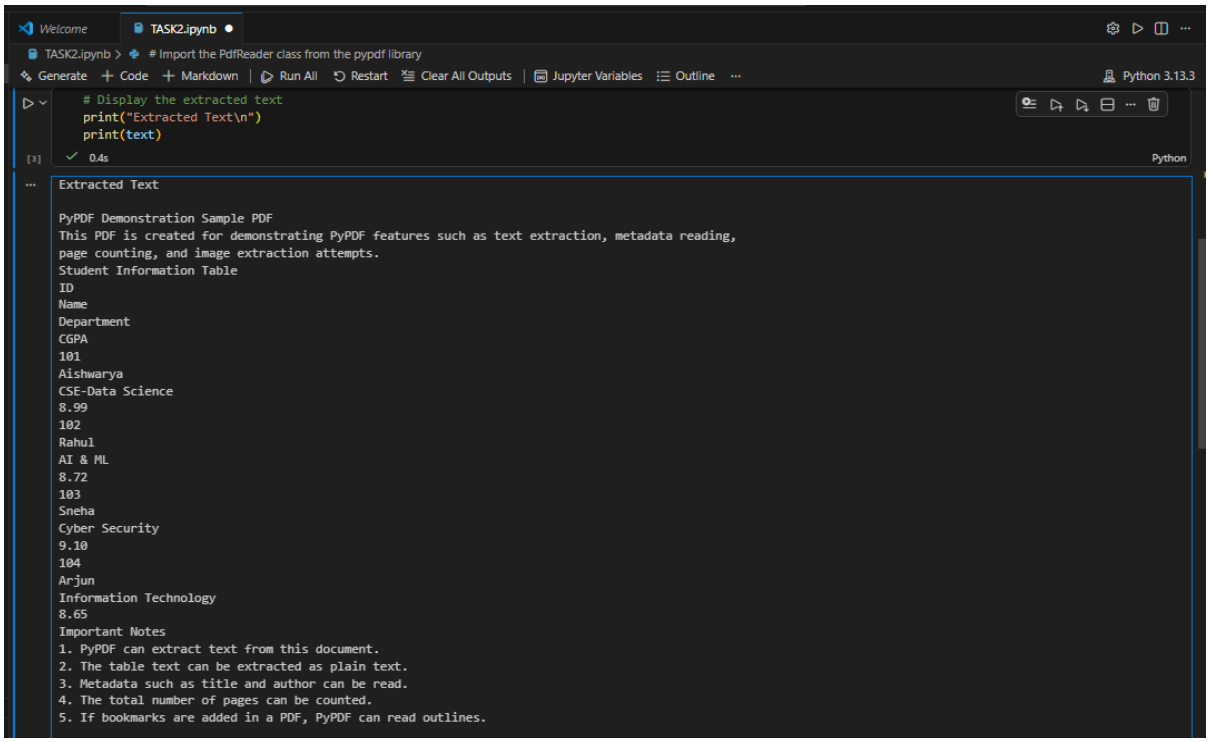
- **PDF that is used for the operations :**



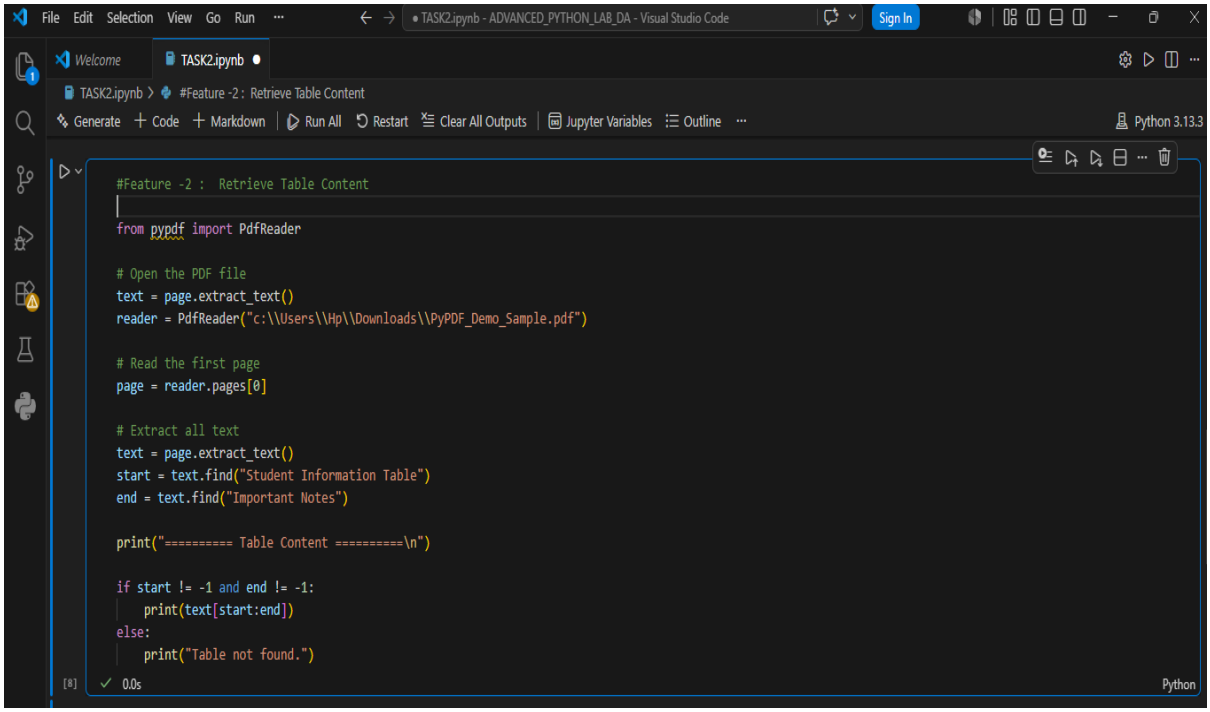
Feature -1 : Extract Text from a PDF



OUTPUT :



Feature -2 : Retrieve Table Content

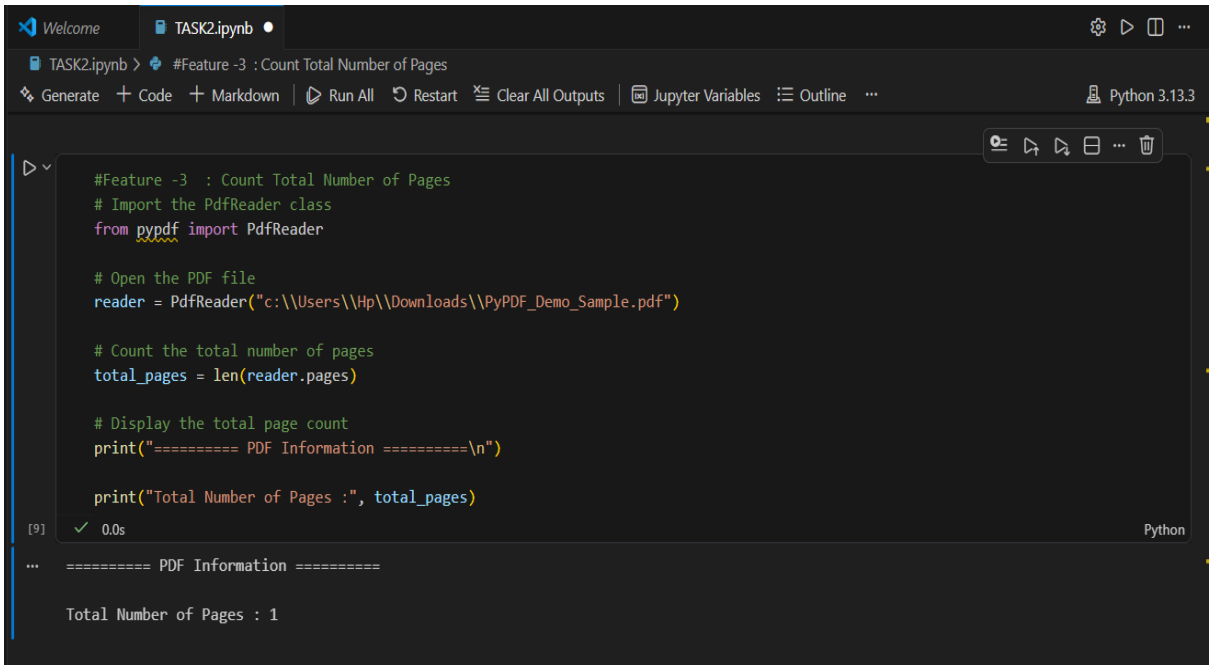


OUTPUT :

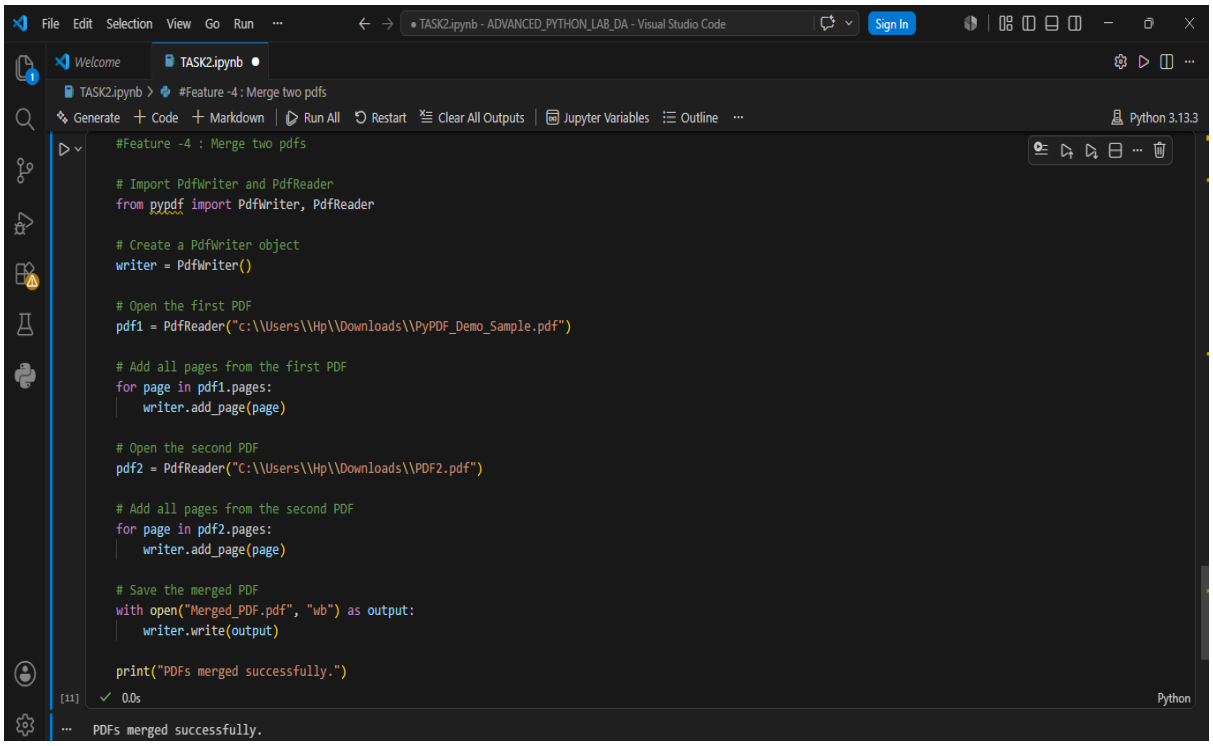
```
========== Table Content ==========

Student Information Table
ID
Name
Department
CGPA
101
Aishwarya
CSE-Data Science
8.99
102
Rahul
AI & ML
8.72
103
Sneha
Cyber Security
9.10
104
Arjun
Information Technology
8.65
```

Feature 3 : Count Total Number of Pages



Feature -4 : Merging two PDFS.



```
#Feature -4 : Merge two pdfs

# Import PdfWriter and PdfReader
from pypdf import PdfWriter, PdfReader

# Create a PdfWriter object
writer = PdfWriter()

# Open the first PDF
pdf1 = PdfReader("c:\\Users\\Hp\\Downloads\\PyPDF_Demo_Sample.pdf")

# Add all pages from the first PDF
for page in pdf1.pages:
    writer.add_page(page)

# Open the second PDF
pdf2 = PdfReader("C:\\Users\\Hp\\Downloads\\PDF2.pdf")

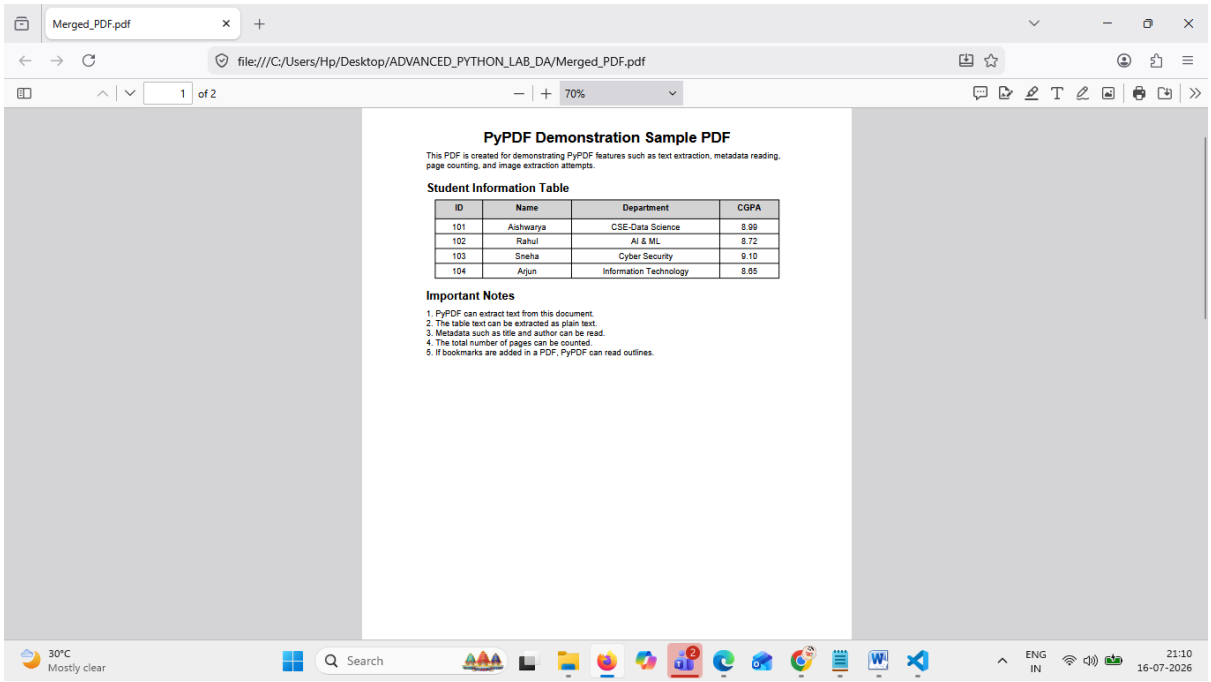
# Add all pages from the second PDF
for page in pdf2.pages:
    writer.add_page(page)

# Save the merged PDF
with open("Merged_PDF.pdf", "wb") as output:
    writer.write(output)

print("PDFs merged successfully.")
```

```
PDFs merged successfully.
```

OUTPUT :



Merged_PDF.pdf

file:///C:/Users/Hp/Desktop/ADVANCED_PYTHON_LAB_DA/Merged_PDF.pdf

1 of 2 — 70%

PyPDF Demonstration Sample PDF

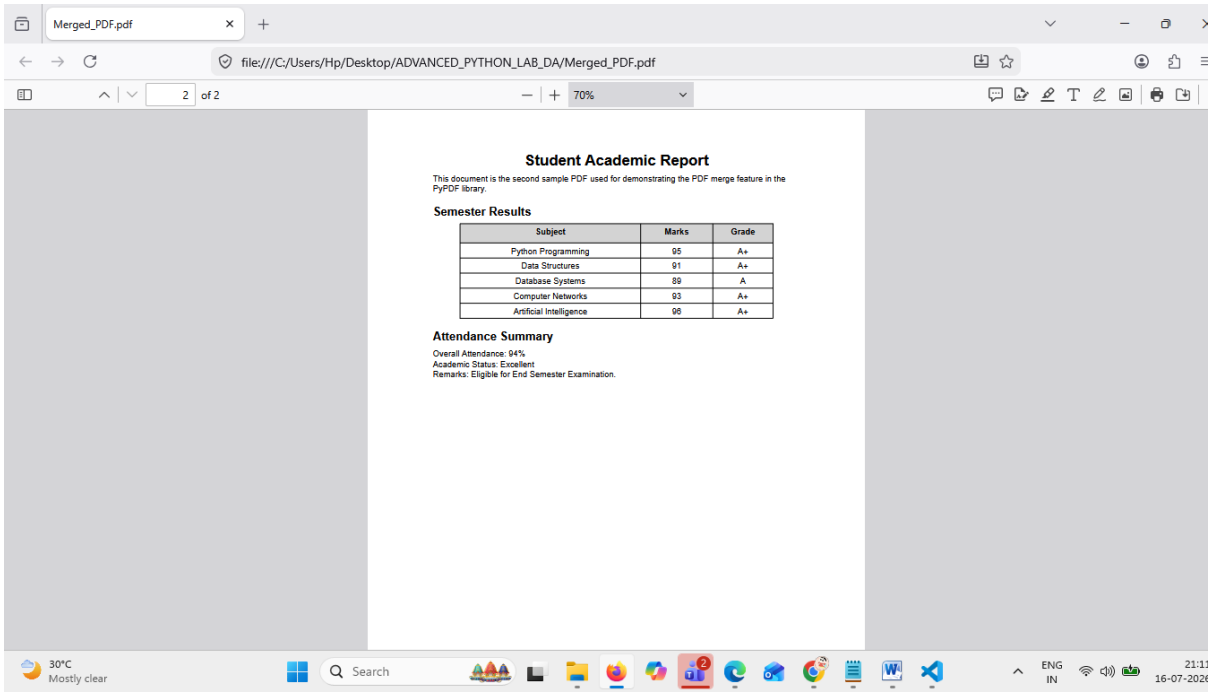
This PDF is created for demonstrating PyPDF features such as text extraction, metadata reading, page counting, and image extraction attempts.

Student Information Table

ID	Name	Department	CGPA
101	Aishwarya	CSE-Data Science	8.99
102	Rahul	AI & ML	8.72
103	Sneha	Cyber Security	9.10
104	Arjun	Information Technology	8.65

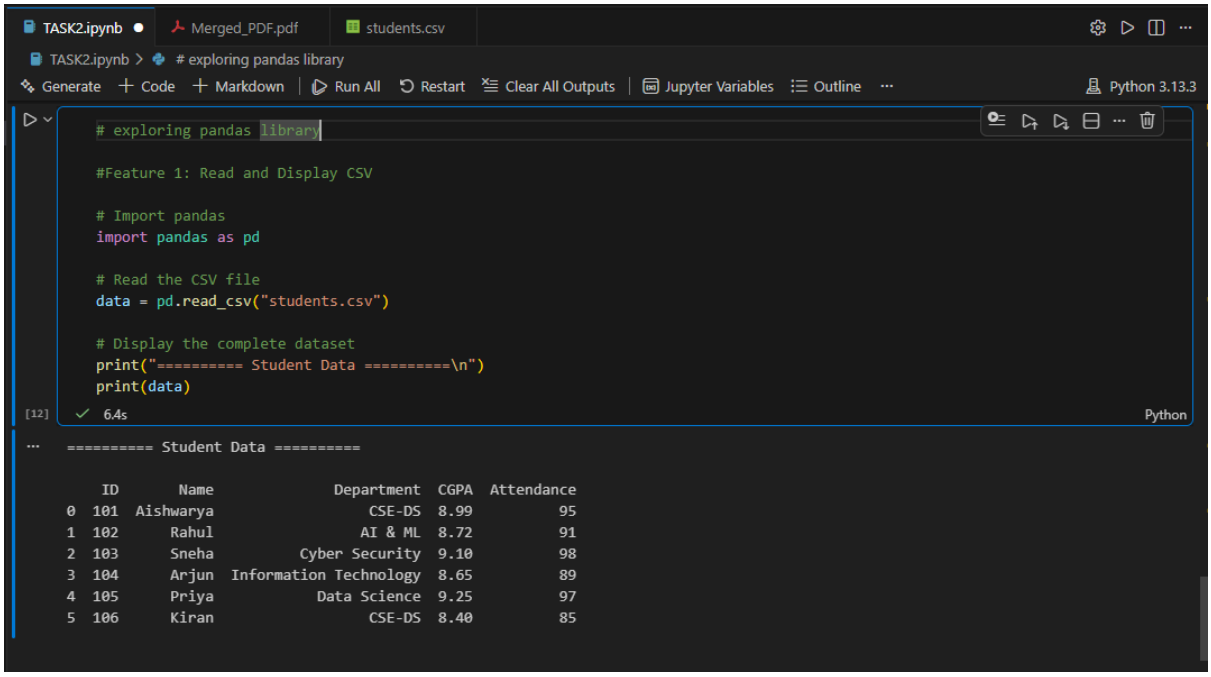
Important Notes

1. PyPDF can extract text from this document.
2. The table text can be extracted as plain text.
3. Metadata such as title and author can be read.
4. The total number of pages can be counted.
5. If bookmarks are added in a PDF, PyPDF can read outlines.

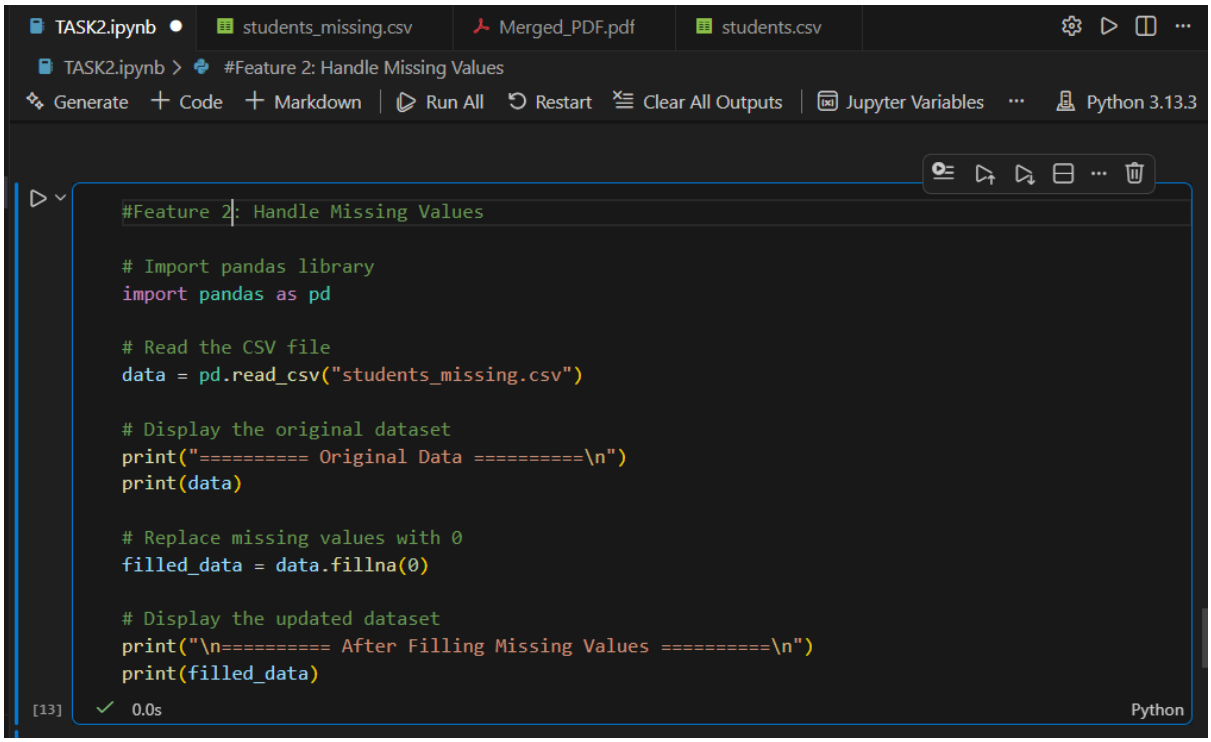


2.pandas for .csv files.

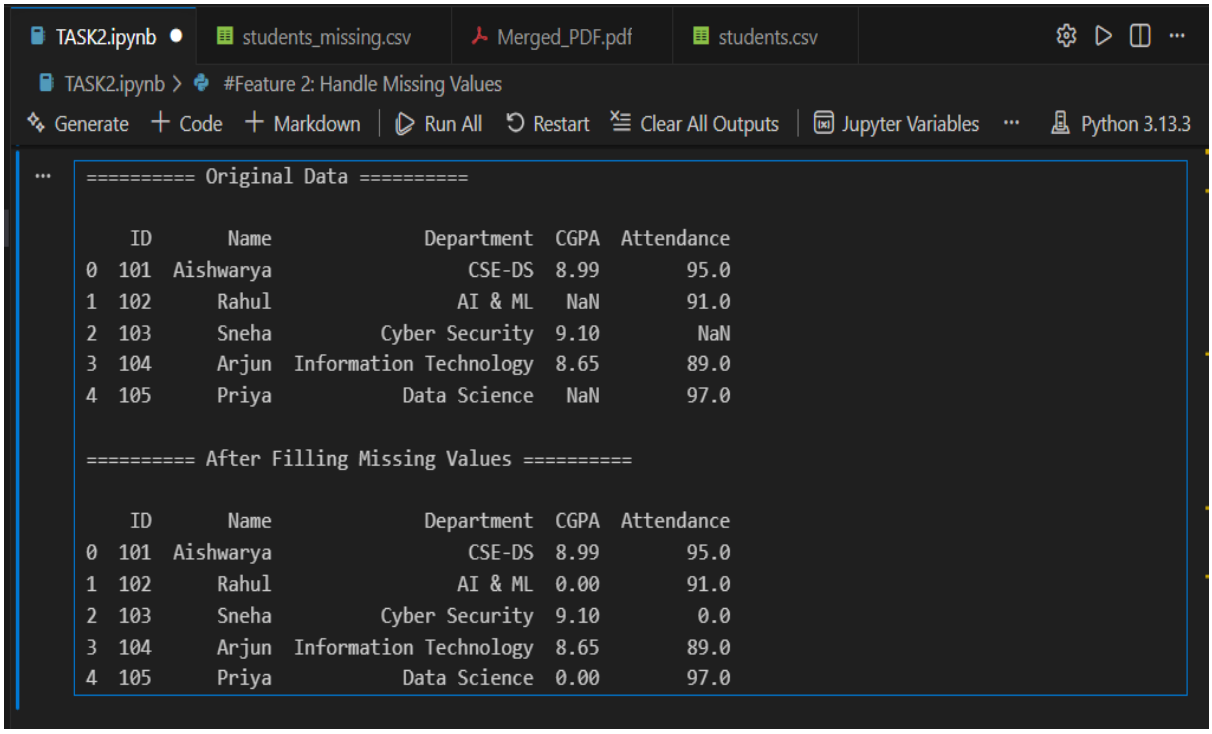
Feature 1:For Reading and Displaying CSV data.



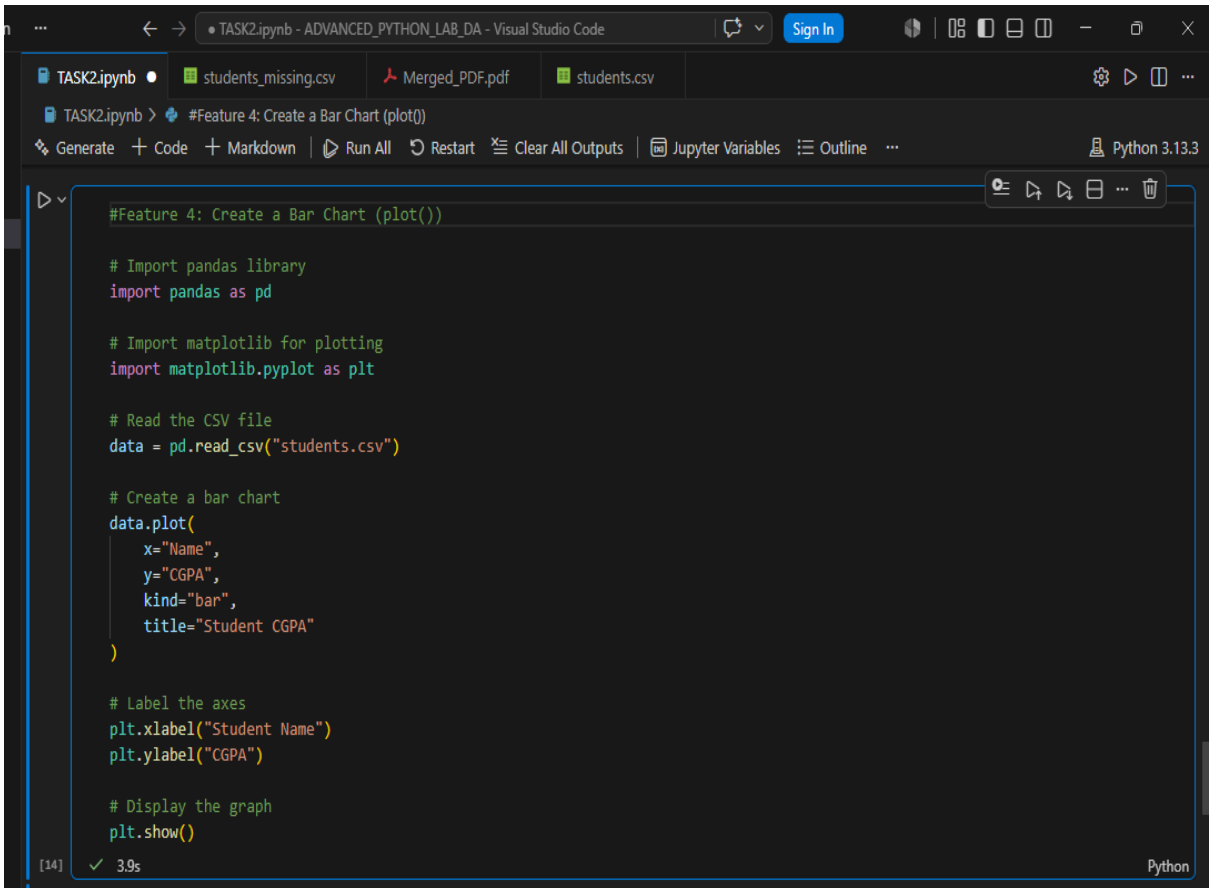
Feature 2: Handle Missing Values of csv datasets.



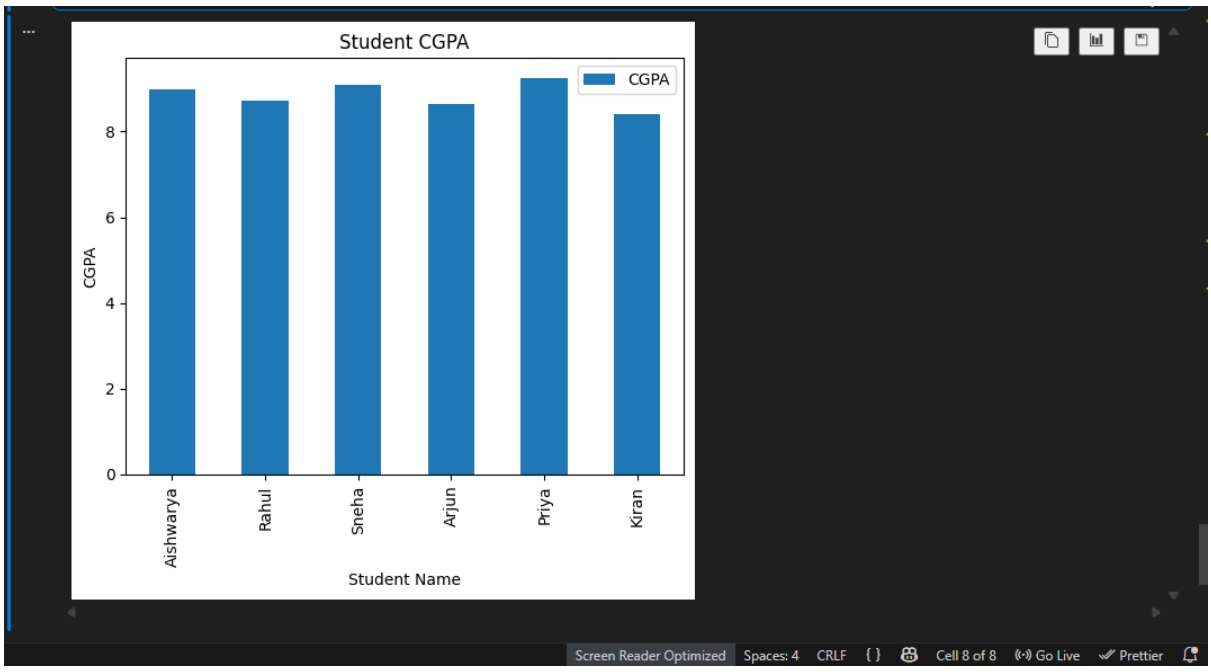
OUTPUT :



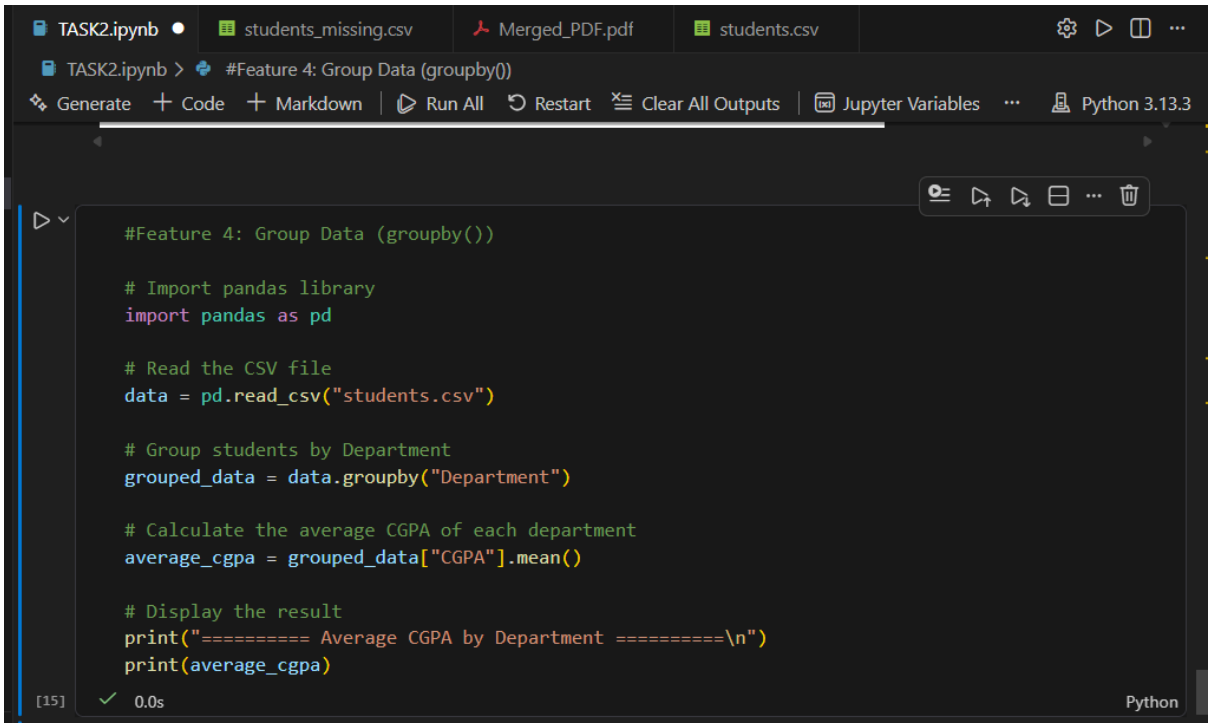
Feature 3: Creating visualisations (plot())



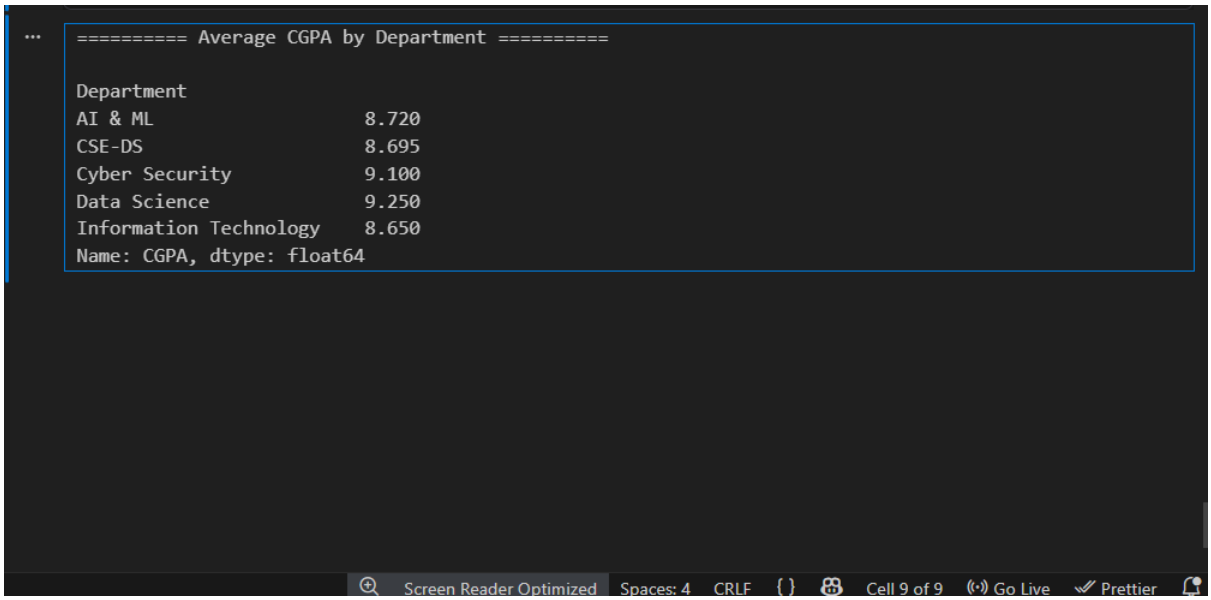
OUTPUT :



Feature 4 : Grouping Data (groupby())



OUTPUT :

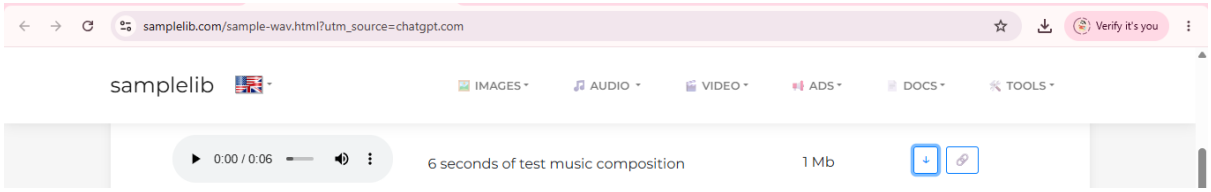


TASK- 3 : DATA FORMAT OF AUDIO

To load an audio file and convert it into analog format and then apply a low-pass filter, and analyze its amplitude and dominant frequency using Python.

Steps :

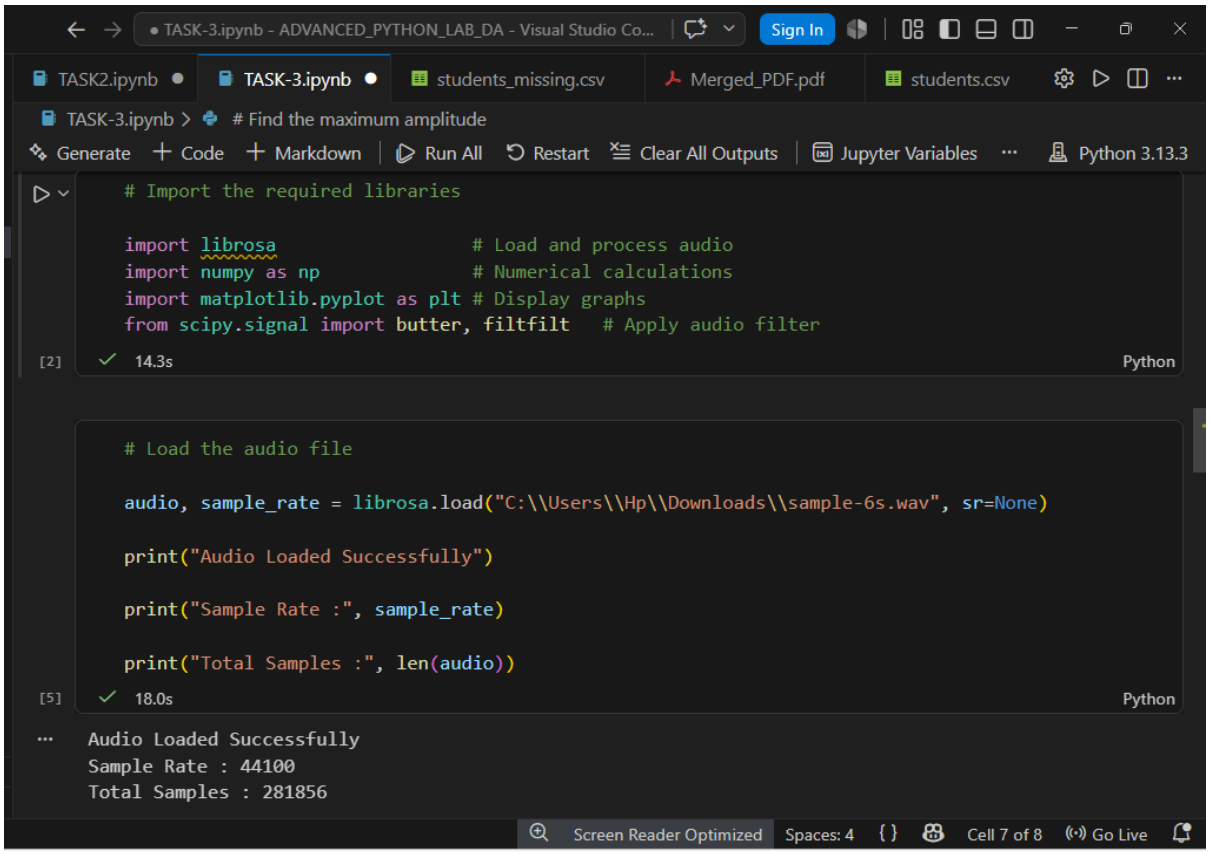
- Download an audio file



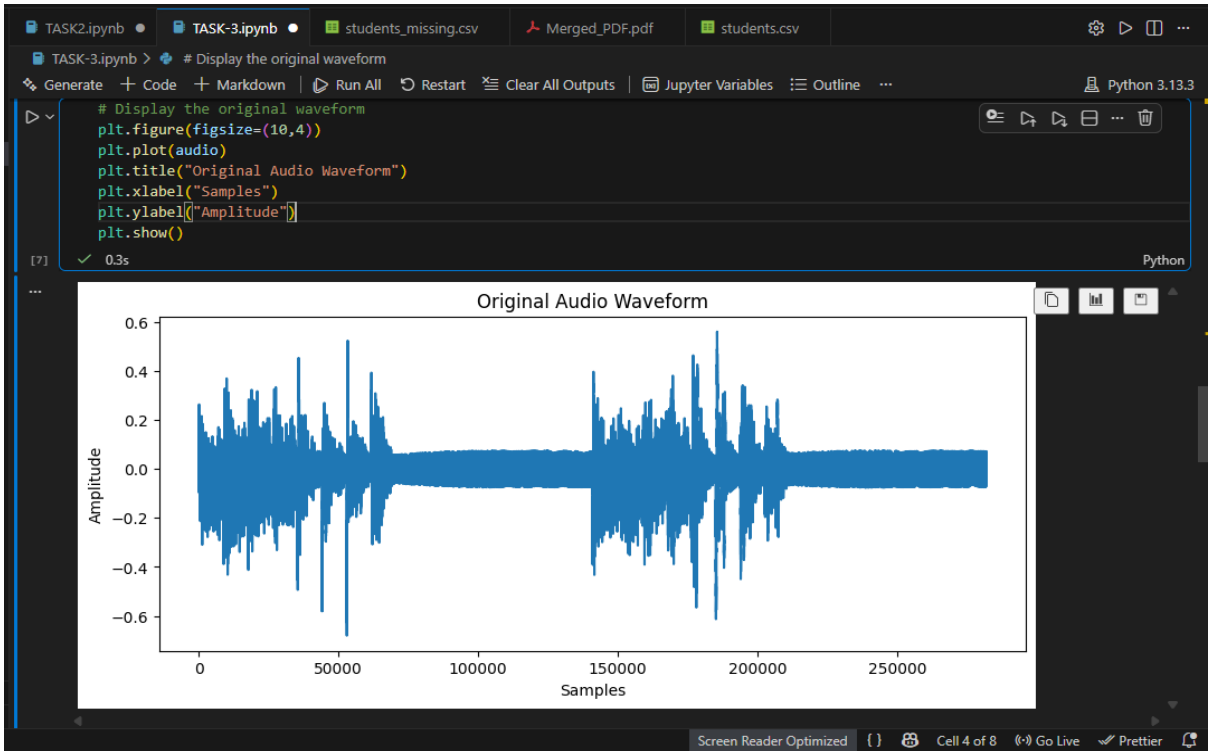
STEPS :

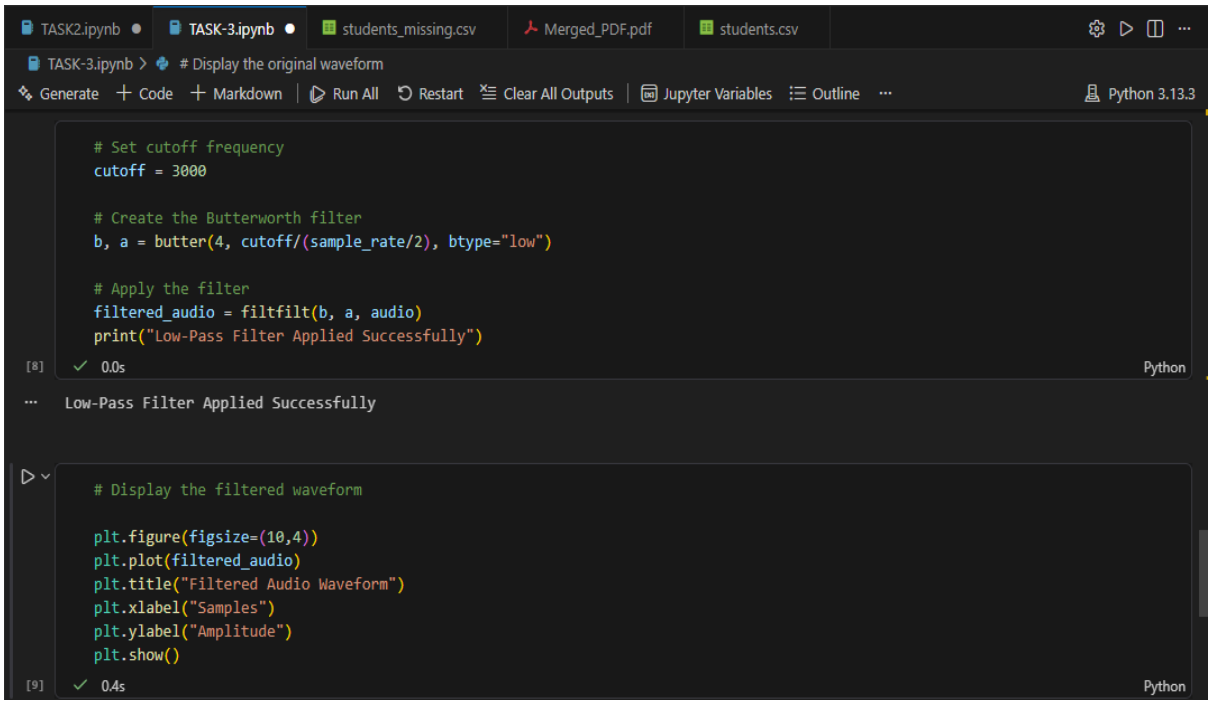
1: Import Required Libraries

2: Load the Audio File

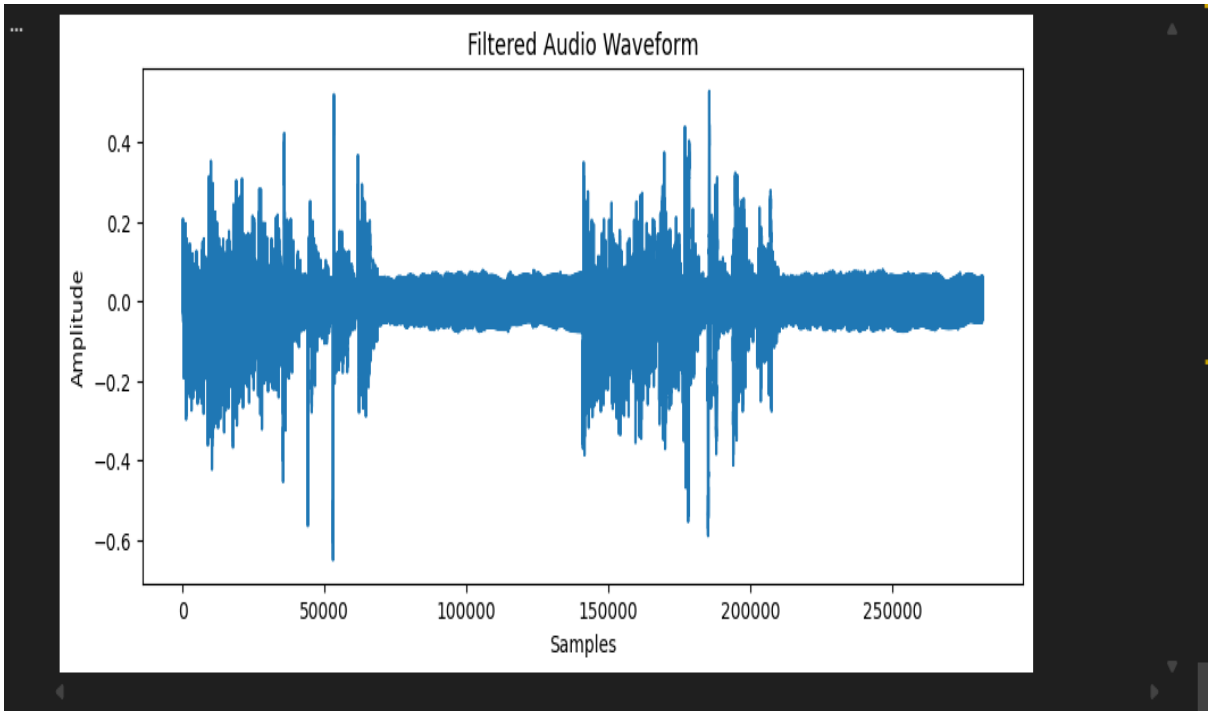


3: Display the Original Audio Waveform



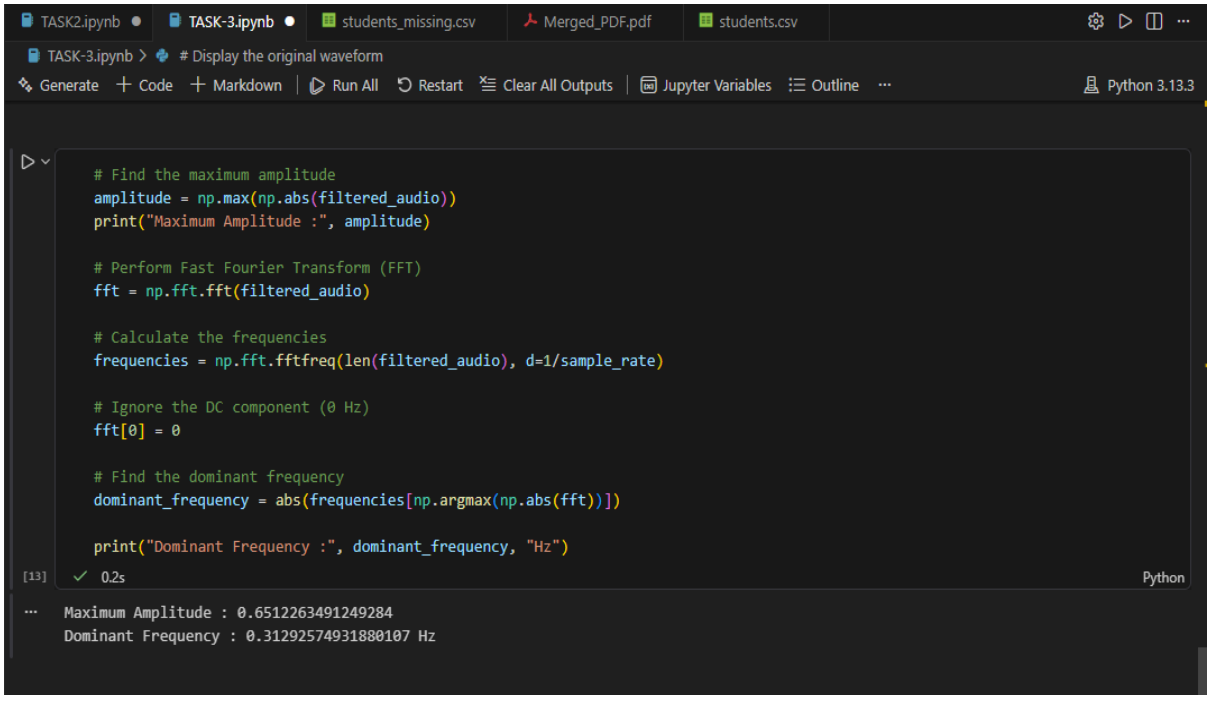


Step 4: Display the Filtered Audio Waveform



6: Calculate the Maximum Amplitude

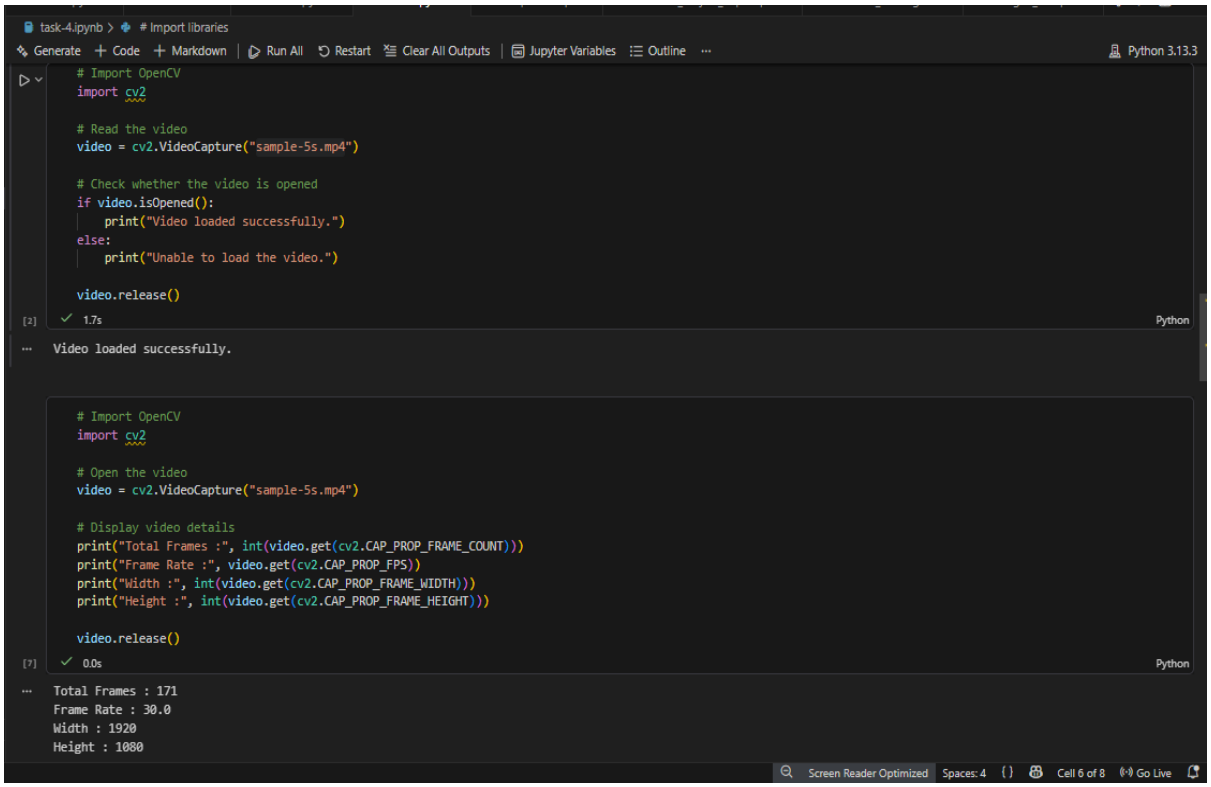
7: Perform Frequency Analysis Using FFT



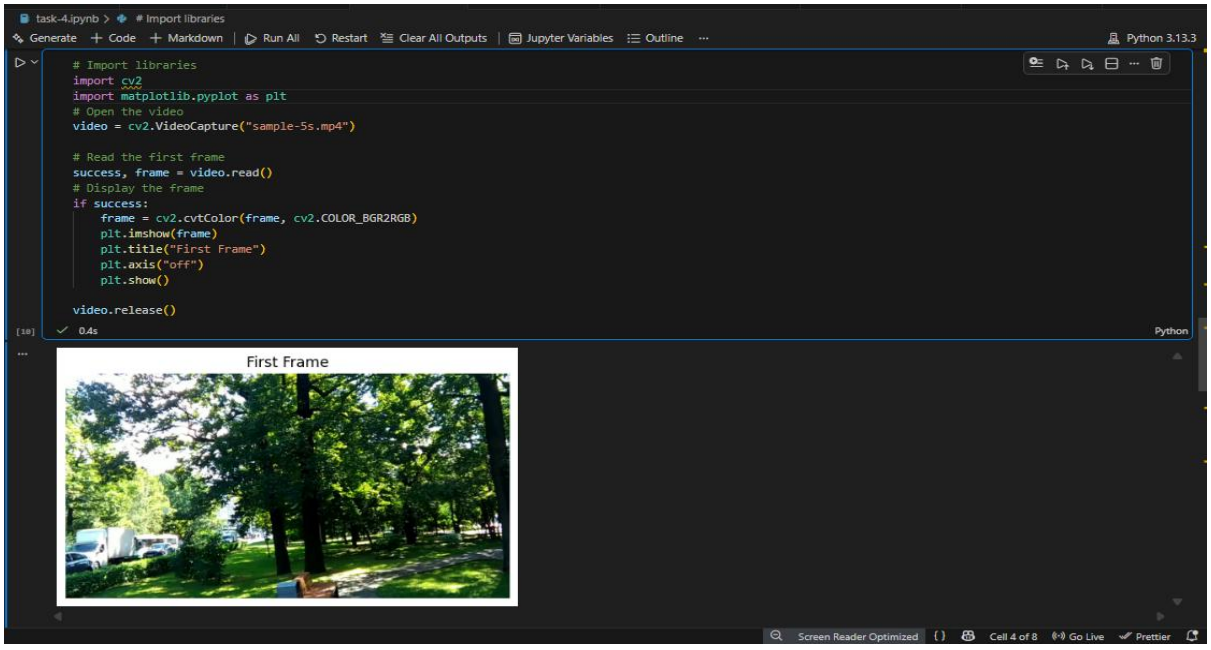
TASK 4: DATA FORMAT OF VIDEO

To read a video file, extract the first frame and analyze the amplitude and frequency using Python.

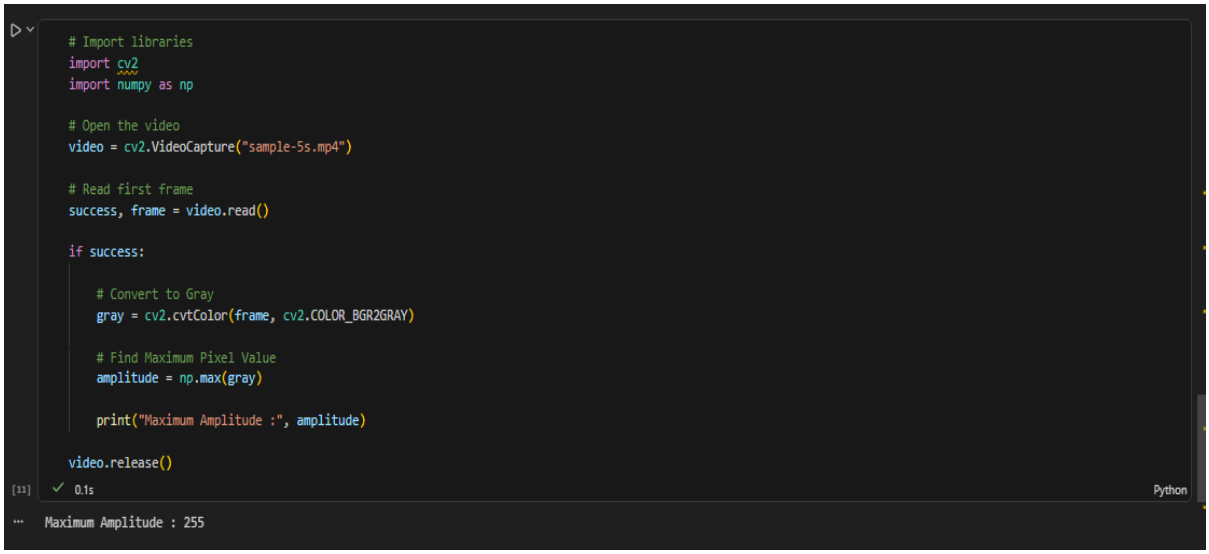
1. **Load the Video File**
2. **Display the Video Information**



3.Extract the First Frame



4.Calculate the Maximum Amplitude



```python
# Import libraries
import cv2
import numpy as np

# Open the video
video = cv2.VideoCapture("sample-5s.mp4")

# Read first frame
success, frame = video.read()

if success:

    # Convert to Gray
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Find Maximum Pixel Value
    amplitude = np.max(gray)

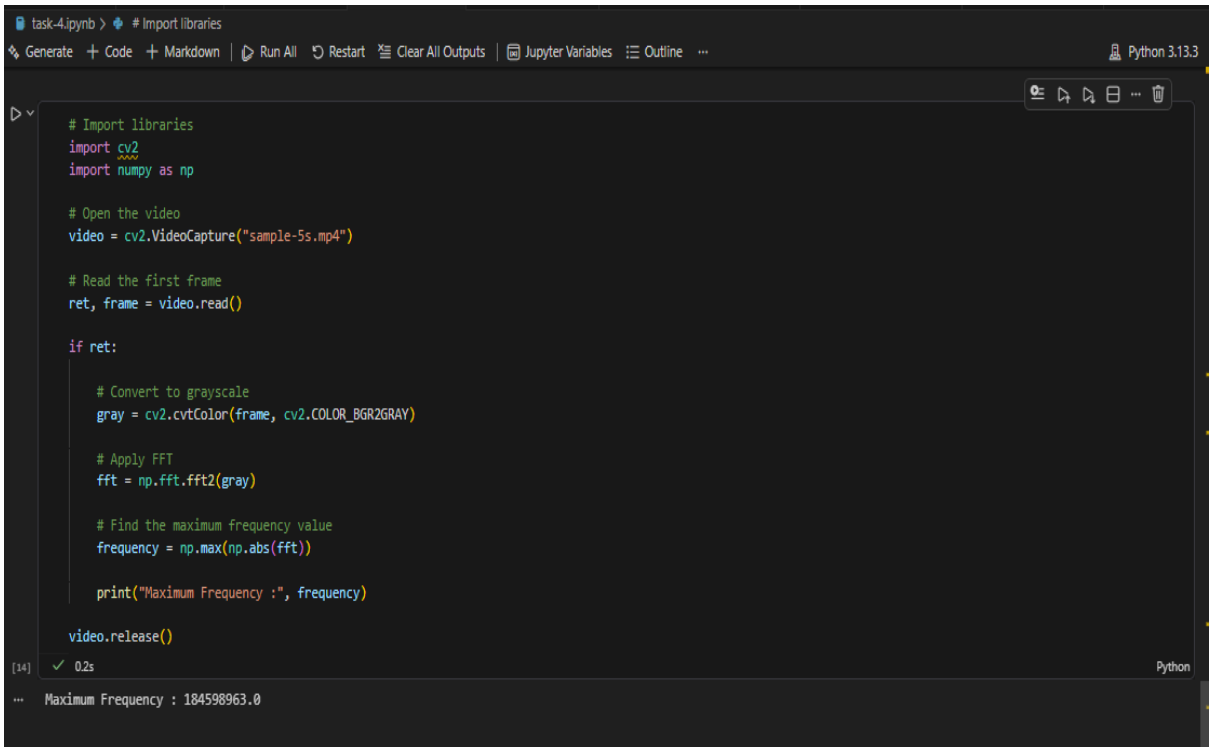
    print("Maximum Amplitude :", amplitude)

video.release()
```

[11] ✓ 0.1s Python

```
Maximum Amplitude : 255
```

5.Perform Frequency Analysis



task-4.ipynb > # Import libraries

Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline ... Python 3.13.3

```python
# Import libraries
import cv2
import numpy as np

# Open the video
video = cv2.VideoCapture("sample-5s.mp4")

# Read the first frame
ret, frame = video.read()

if ret:

    # Convert to grayscale
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Apply FFT
    fft = np.fft.fft2(gray)

    # Find the maximum frequency value
    frequency = np.max(np.abs(fft))

    print("Maximum Frequency :", frequency)

video.release()
```

[14] ✓ 0.2s Python

```
Maximum Frequency : 184598963.0
```

Task 5: Scraping a Webpage

To access a webpage, verify that it is accessible, extract the webpage content, and calculate the total number of words and characters using Python.

```python
# Import required libraries
import requests
from bs4 import BeautifulSoup
import textwrap

# Enter the webpage URL
url = "https://quotes.toscrape.com"

# Send request
response = requests.get(url)

# Check if webpage is accessible
if response.status_code == 200:

    print("Webpage loaded successfully.\n")

    # Read webpage content
    soup = BeautifulSoup(response.text, "html.parser")

    content = soup.get_text(separator=" ", strip=True)

    print("Webpage Content:\n")

    # Display text neatly (80 characters per line)
    print(textwrap.fill(content, width=80))

    # Count words and characters
    print("\nWord Count :", len(content.split()))
    print("Character Count :", len(content))

else:
    print("Unable to load the webpage.")
```

[26] ✓ 1.0s Python

OUTPUT :

```
Webpage loaded successfully.

Webpage Content:

Quotes to Scrape Quotes to Scrape Login “The world as we have created it is a
process of our thinking. It cannot be changed without changing our thinking.” by
Albert Einstein (about) Tags: change deep-thoughts thinking world “It is our
choices, Harry, that show what we truly are, far more than our abilities.” by
J.K. Rowling (about) Tags: abilities choices “There are only two ways to live
your life. One is as though nothing is a miracle. The other is as though
everything is a miracle.” by Albert Einstein (about) Tags: inspirational life
live miracle miracles “The person, be it gentleman or lady, who has not pleasure
in a good novel, must be intolerably stupid.” by Jane Austen (about) Tags:
aliteracy books classic humor “Imperfection is beauty, madness is genius and
it's better to be absolutely ridiculous than absolutely boring.” by Marilyn
Monroe (about) Tags: be-yourself inspirational “Try not to become a man of
success. Rather become a man of value.” by Albert Einstein (about) Tags:
adulthood success value “It is better to be hated for what you are than to be
loved for what you are not.” by André Gide (about) Tags: life love “I have not
failed. I've just found 10,000 ways that won't work.” by Thomas A. Edison
(about) Tags: edison failure inspirational paraphrased “A woman is like a tea
bag; you never know how strong it is until it's in hot water.” by Eleanor
Roosevelt (about) Tags: misattributed-eleanor-roosevelt “A day without sunshine
is like, you know, night.” by Steve Martin (about) Tags: humor obvious simile
Next → Top Ten tags love inspirational life humor books reading friendship
friends truth simile Quotes by: GoodReads.com Made with ♥ by Zyte

Word Count : 282
Character Count : 1681
```

THANKYOU